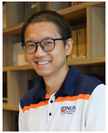
Active Clustering with Noisy Pairwise Observations

Karthik P N

SPARC Workshop on Distributed Learning and Optimization, IISc

13 July 2026

Based on Joint Work With



Rachel Teo (NUS) **Vincent Tan (NUS)** **Ramya Vinayak (UW Madison)**

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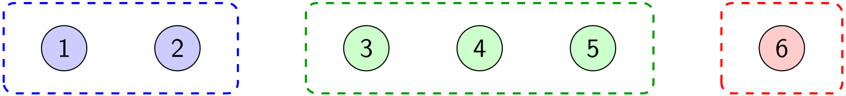
Problem Setting



- M items, each belonging to one of $K \in [M]$ clusters; K unknown

  Example: $M = 6$; $K = 3$; $c_1 = c_2 = 1$, $c_3 = c_4 = c_5 = 2$, $c_6 = 3$

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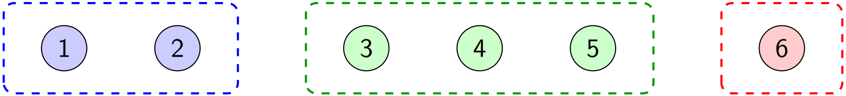

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- At each time $t = 1, 2, \ldots,$ query item pair $(i_t, j_t) = (i, j)$ to get

$$\mathbf{Y}_{ij} \sim v_{ij} = \begin{cases} \mathrm{Ber}(p), & c_i = c_j \\ \mathrm{Ber}(q), & c_i \neq c_j, \end{cases}$$

  where $0 \leq q < 1/2 < p \leq 1$; $\boxed{p, q \text{ unknown}}$

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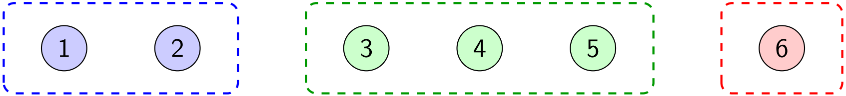

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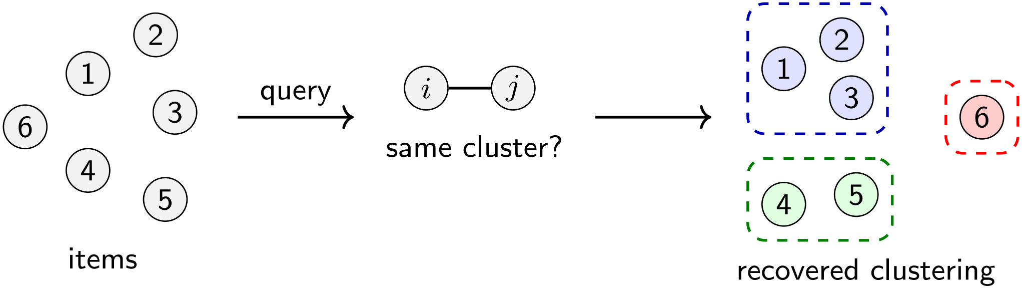
- **Goal**: Given $\delta \in (0, 1)$, recover the underlying clustering correctly w.p. $\geq 1 - \delta$


1
2
3
4
5
6
items
query
i
j
same cluster?
1
2
3
6
4
5
recovered clustering

Instance Matrix and Adjacency Matrix

Instance C

	1	2	3	4	5	6
1	p	p	q	q	q	q
2	p	p	q	q	q	q
3	q	q	p	p	p	q
4	q	q	p	p	p	q
5	q	q	p	p	p	q
6	q	q	q	q	q	p

Adjacency Matrix $C^{=}$

	1	2	3	4	5	6
1	1	1	0	0	0	0
2	1	1	0	0	0	0
3	0	0	1	1	1	0
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$C^{\neq}$

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$$C = p\,C^{=} + q\,C^{\neq}$$

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- The closer p and q are, the harder it is to recover $C^{=}$

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$$C = p\,C^{=} + q\,C^{\neq}$$

- Our interest is only to recover $C^{=}$, not the instance matrix C
- The closer p and q are, the harder it is to recover $C^{=}$
- Two instances are equivalent if their adjacency matrices are identical

$$C' \sim C'' \iff C'^{=} = C''^{=}.$$

Instance Space and Equivalence Classes

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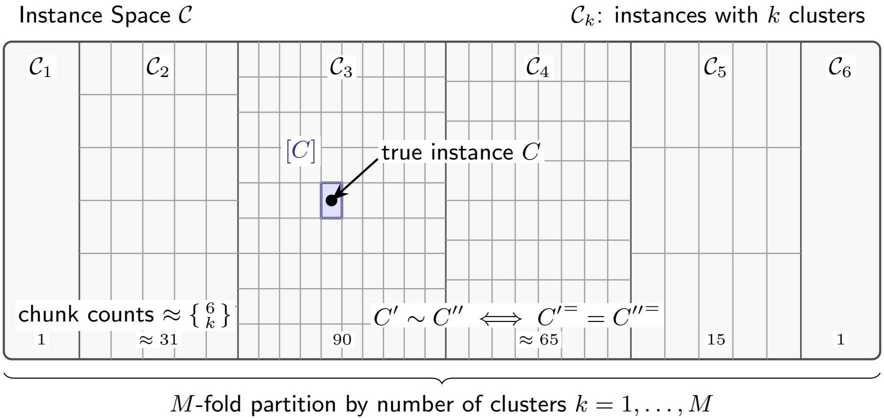

$$\mathcal{C} = \bigcup_{k=1}^{M} \mathcal{C}_k,$$

$$\#\text{chunks in } \mathcal{C}_k = \left\{ {M \atop k} \right\}$$

Stirling number of the second kind

Active Clustering vs SBM

	Active Clustering (Our Setup)	Stochastic Block Model (SBM)
Observations	Actively query item pairs and receive noisy binary feedback; the same pair may be queried multiple times	Observe one random graph snapshot; each potential edge is sampled once through graph connectivity
Asymptotics	M fixed, $\delta \downarrow 0$ to zero	$M \to \infty$
Probabilities	p, q fixed and unknown	p, q scale like $\Theta\left(\frac{\log M}{M}\right)$

Takeaway: Ours is an adaptive, fixed-confidence sampling problem rather than passive recovery from a single random graph

Alternative Set

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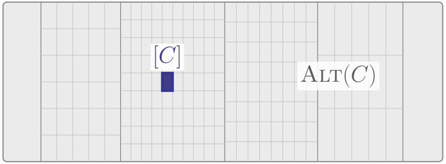
- Underlying instance: C (unknown)
- The set of alternative instances is

$$\textsc{Alt}(C) = \bigcup_{\substack{C' \in \mathcal{C}: \\ [C'] \neq [C]}} [C'].$$



Theorem (Instance-Dependent Lower Bound)

Any δ-correct algorithm π satisfies

$$\mathbb{E}_C[\tau_\delta(\pi)] \geq \frac{\mathrm{KL}\big(\mathrm{Ber}(\delta) \parallel \mathrm{Ber}(1-\delta)\big)}{D^\star(C)},$$

where

$$D^\star(C) = \sup_{\lambda \in \mathcal{P}_\mathcal{I}} \inf_{C' \in \textsc{Alt}(C)} \sum_{(i,j) \in \mathcal{I}} \lambda_{ij} \, \mathrm{KL}(v_{ij}, v'_{ij}).$$

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- λ is the algorithm's sampling strategy
- Learner: choose λ to maximize the information gathered

Lower Bound

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- For a fixed λ, Nature chooses the most confusing alternative C'
- The inner $\inf$ identifies the hardest alternative

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- Weighted KL sum measures the distinguishability of C' from C under λ
- Learner vs. Nature: a max–min game whose value is $D^\star(C)$

A Key Reduction

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Theorem (Reduction to Minimal Alternatives)

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$$D^\star(C) = \sup_{\lambda \in \mathcal{P}_I} \inf_{C' \in \textsc{Alt}(C)} \sum_{(i,j)\in I} \lambda_{ij} \mathrm{KL}(v_{ij}, v'_{ij})$$

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- $\textsc{Min}(C)$: set of alternatives obtained by splitting a cluster in C into two clusters or merging two clusters in C into one cluster

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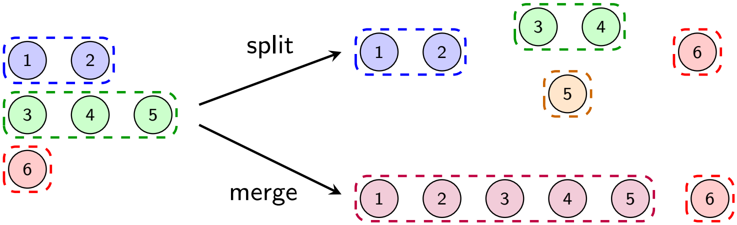

$\textsc{Alt}(C) \;\rightarrow\; \textsc{Min}(C)$

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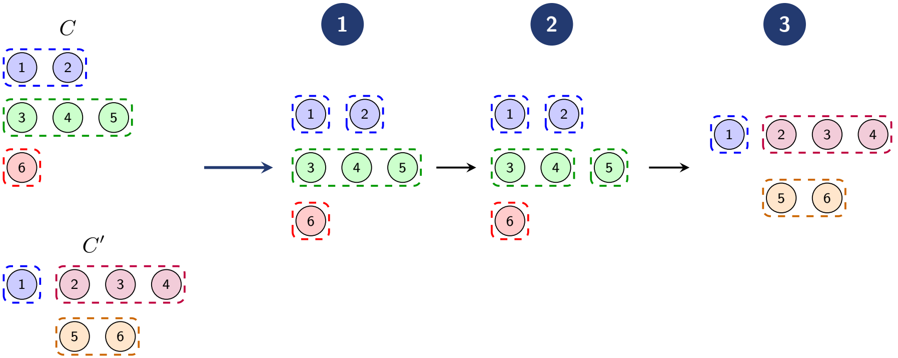

- $N_1(C, C') = \{(i,j) : \;\; c_i = c_j,\; c'_i \neq c'_j\} \;\; = \;\; \{(1,2), (3,5), (4,5)\}$
  $N_2(C, C') = \{(i,j) : \;\; c_i \neq c_j,\; c'_i = c'_j\} \;\; = \;\; \{(2,3), (2,4), (5,6)\}$

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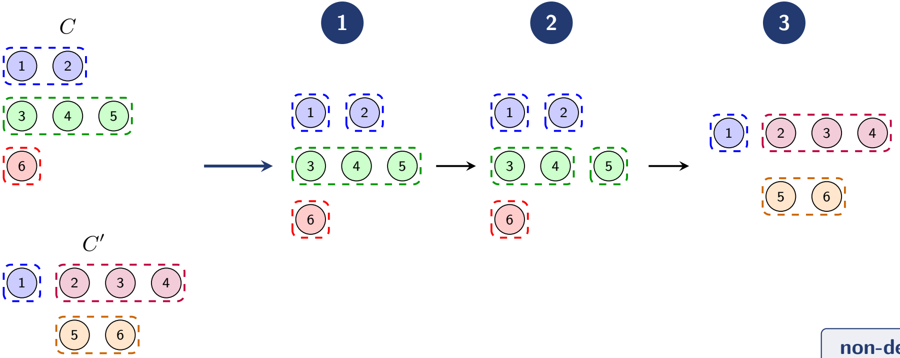

- $N_1(C,C') = \{(i,j):\; c_i = c_j,\; c'_i \neq c'_j\} \;=\; \{(1,2),(3,5),(4,5)\}$
  $N_2(C,C') = \{(i,j):\; c_i \neq c_j,\; c'_i = c'_j\} \;=\; \{(2,3),(2,4),(5,6)\}$
- The inner infimum can be written as

non-decreasing

$$\inf_{C' \in \textsc{Alt}(C)} \sum_{(i,j)\in I} \lambda_{ij}\; \mathrm{KL}(v_{ij}, v'_{ij}) \;=\; \min_{[C'] \in \textsc{Alt}(C)/\sim} z\big(\lambda_{N_1(C,C')},\; \lambda_{N_2(C,C')}\big)\,.$$

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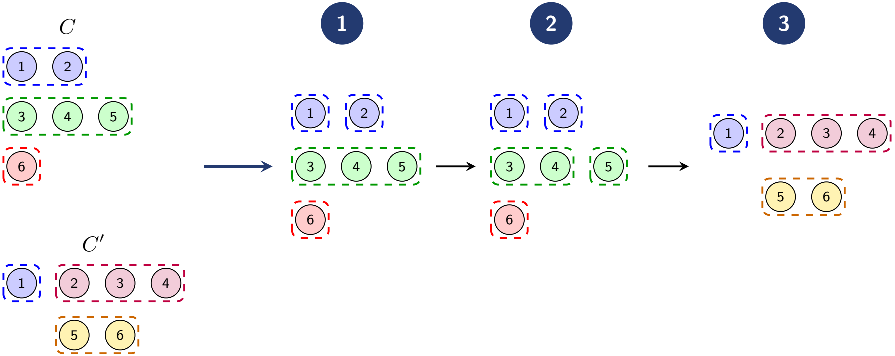

- $N_1(C, C') = \{(1,2), (3,5), (4,5)\}, \qquad N_1(C, 1) = \{(1,2)\}$
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- $\lambda_{N_1(C,C')} = \lambda_{12} + \lambda_{35} + \lambda_{45}, \qquad \lambda_{N_1(C,1)} = \lambda_{12}$
  $\lambda_{N_2(C,C')} = \lambda_{23} + \lambda_{24} + \lambda_{56}, \qquad \lambda_{N_2(C,1)} = 0$

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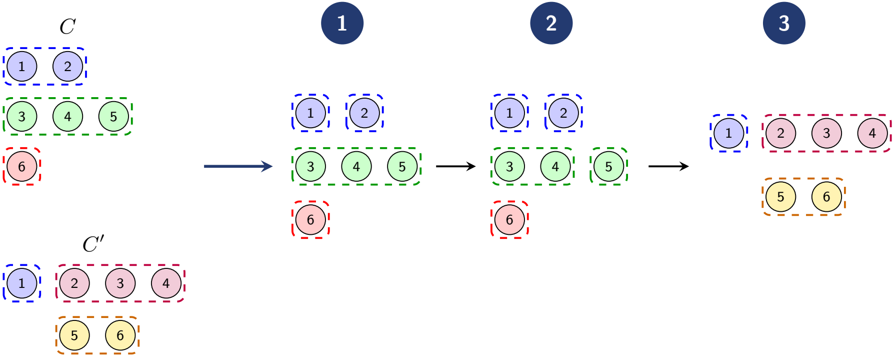

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Algorithm: Sampling Rule

- Compute the empirical instance $\widehat{C}_t$ given by

$$(\widehat{C}_t)_{ij} \;=\; (\widehat{C}_t)_{ji} \;=\; \frac{1}{N_{ij}(t)} \sum_{s=1}^{t} \mathbb{1}\{i_s = i,\ j_s = j\}\, \mathbf{Y}_{i_s j_s},$$

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- Compute the set of optimal allocations for $\widehat{C}_t$:

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- $\widehat{C}_t \notin \mathcal{C}$

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- Sample (i_{t+1}, j_{t+1}) according to any $\lambda \in \Lambda^\star(\widehat{C}_t)$

Why this does not work:

- C has only two levels (p,q), but $\widehat{C}_t$ need not
- $\widehat{C}_t \notin \mathcal{C}$, and hence $\Lambda^\star(\widehat{C}_t)$ is not well-defined

Empirical pairwise means

$$
\begin{aligned}
(\widehat{C}_t)_{ij} &= (\widehat{C}_t)_{ji} \\
&= \frac{1}{N_{ij}(t)} \sum_{s=1}^{t} \mathbb{1}\{i_s = i,\ j_s = j\}\, \mathbf{Y}_{i_s j_s}
\end{aligned}
$$

Typical empirical instance $\widehat{C}_t$

	1	2	3	4	5	6
1	1	.78	.34	.22	.47	.18
2	.78	1	.41	.29	.36	.53
3	.34	.41	1	.72	.61	.44
4	.22	.29	.72	1	.67	.38
5	.47	.36	.61	.67	1	.27
6	.18	.53	.44	.38	.27	1

Many empirical levels, not just p, q

Solution: Project $\widehat{C}_t$ onto $\mathcal{C}$ using $\mathrm{Proj}_{\mathcal{C}}$, and set $C_t = \mathrm{Proj}_{\mathcal{C}}(\widehat{C}_t)$.

Empirical pairwise means

$$(\widehat{C}_t)_{ij} = (\widehat{C}_t)_{ji} = \frac{1}{N_{ij}(t)} \sum_{s=1}^{t} \mathbb{1}\{i_s = i,\ j_s = j\}\, \mathbf{Y}_{i_s j_s}$$

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Solution: Project $\widehat{C}_t$ onto $\mathcal{C}$ using $\mathrm{Proj}_{\mathcal{C}}$, and set $C_t = \mathrm{Proj}_{\mathcal{C}}(\widehat{C}_t)$.

Step 1: Threshold at $1/2$

$$(\widehat{C}_t^{=})_{ij} \quad = \quad \mathbb{1}\left\{(\widehat{C}_t)_{ij} \geq \frac{1}{2}\right\},$$

$$(\widehat{C}_t^{\neq})_{ij} \quad = \quad \mathbb{1}\left\{(\widehat{C}_t)_{ij} < \frac{1}{2}\right\}.$$

for all $(i,j) \in I$.

Empirical matrix $\widehat{C}_t$

	1	2	3	4	5	6
1	1	.78	.34	.22	.47	.18
2	.78	1	.41	.29	.36	.53
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6	.18	.53	.44	.38	.27	1

After thresholding: $\widehat{C}_t^{=}$

	1	2	3	4	5	6
1	1	1	0	0	0	0
2	1	1	0	0	0	1
3	0	0	1	1	1	0
4	0	0	1	1	1	0
5	0	0	1	1	1	0
6	0	1	0	0	0	1

$\widehat{C}_t^{\neq} = \mathbf{1} - \widehat{C}_t^{=}$ entrywise

Solution: Project $\widehat{C}_t$ onto $\mathcal{C}$ using $\mathrm{Proj}_{\mathcal{C}}$, and set $C_t = \mathrm{Proj}_{\mathcal{C}}(\widehat{C}_t)$.

Step 2: Find the Nearest Clustering to $\widehat{C}_t^{=}$

$$\widetilde{C}_t = \operatorname*{arg\,min}_{M \in \mathcal{C}/\sim} \sum_{(i,j)\in I} \left|(\widehat{C}_t^{=})_{ij} - M_{ij}^{=}\right| + \left|(\widehat{C}_t^{\neq})_{ij} - M_{ij}^{\neq}\right|$$

Empirical matrix $\widehat{C}_t$

	1	2	3	4	5	6
1	1	.78	.34	.22	.47	.18
2	.78	1	.41	.29	.36	.53
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Step 3: Estimate p_t and q_t

Entries at least 1/2

$$p_t = \frac{\sum_{(i,j)\in I} \mathbb{1}\left\{(\widehat{C}_t^{=})_{ij} = 1\right\} (\widehat{C}_t)_{ij} N_{ij}(t)}{\sum_{(i,j)\in I} \mathbb{1}\left\{(\widehat{C}_t^{=})_{ij} = 1\right\} N_{ij}(t)}.$$

Entries below 1/2

$$q_t = \frac{\sum_{(i,j)\in I} \mathbb{1}\left\{(\widehat{C}_t^{\neq})_{ij} = 1\right\} (\widehat{C}_t)_{ij} N_{ij}(t)}{\sum_{(i,j)\in I} \mathbb{1}\left\{(\widehat{C}_t^{\neq})_{ij} = 1\right\} N_{ij}(t)}.$$

Empirical matrix $\widehat{C}_t$

	1	2	3	4	5	6
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	1	2	3	4	5	6
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Step 4: Form the Projected Instance C_t

$$C_t \quad = \quad p_t\, \widetilde{C}_t + q_t\, (\mathbf{1} - \widetilde{C}_t).$$

Empirical matrix $\widehat{C}_t$

	1	2	3	4	5	6
1	1	.78	.34	.22	.47	.18
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$\widehat{C}_t^{\neq} = \mathbf{1} - \widehat{C}_t^{=}$ entrywise

Sampling Rule

- **Unique optimizer:**

$$\lambda^\star(\sigma; C_t) = \operatorname*{arg\,max}_{\lambda \in \mathcal{P}_I} \left\{ \inf_{C' \in \textsc{Min}(C_t)} \sum_{(i,j) \in I} \lambda_{ij} \ \mathrm{KL}((v_t)_{ij}, v'_{ij}) - \frac{\sigma}{2} \|\lambda\|^2 \right\}.$$

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- **Positive exploration:**

$$\lambda^\star_\epsilon(\sigma; C_t) = (1-\epsilon)\, \lambda^\star(\sigma; C_t) + \epsilon \, \mathrm{Unif}(I).$$

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$$(i_{t+1}, j_{t+1}) \in \underset{(i,j)\in I}{\arg\max} \left\{ t\,(\lambda^\star_\epsilon(\sigma; C_t))_{ij} - N_{ij}(t) \right\}.$$

[1] Garivier, A., & Kaufmann, E. (2016). Optimal best arm identification with fixed confidence. In V. Feldman, A. Rakhlin, & O. Shamir (Eds.), *Proceedings of the 29th Annual Conference on Learning Theory* (Vol. 49, pp. 998–1027). PMLR.

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- Consequently,

$$\frac{N(t)}{t} \xrightarrow{\text{a.s.}} \lambda^\star_\epsilon(\sigma; C).$$

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Stopping Rule

- A natural stopping rule is to use the GLR statistic

$$Z(t) = \inf_{C' \in \textsc{Alt}(C_t)} \sum_{(i,j) \in I} N_{ij}(t)\, \mathrm{KL}((\widehat{v}_t)_{ij}, v'_{ij}),$$

and to stop once $Z(t)$ has crossed a threshold $\beta(t, \delta)$:

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Using the threshold in Kaufmann and Koolen (2021)[1],

Theorem (δ-correct Stopping Rule)

The GLR-based stopping rule is δ-correct, i.e.,

$$\Pr\left(\tau_\delta < \infty,\ [C_{\tau_\delta}] \neq [C]\right) \leq \ \delta.$$

[1] Kaufmann, E., & Koolen, W. M. (2021). Mixture martingales revisited with applications to sequential tests and confidence intervals. *Journal of Machine Learning Research*, *22*(246), 1–44.

Theorem (Almost Asymptotically Optimal Algorithm)

Under D-tracking and the GLR-based stopping rule,

$$\limsup_{\delta \to 0} \frac{\mathbb{E}_C[\tau_\delta]}{\ln(1/\delta)} \leq \frac{1}{D^\star_\epsilon(\sigma; C)},$$

where

$$D^\star_\epsilon(\sigma; C) = \inf_{C' \in \textsc{Alt}(C)} \sum_{(i,j) \in I} (\lambda^\star_\epsilon(\sigma; C))_{ij} \ \mathrm{KL}(v_{ij}, v'_{ij}) - \frac{\sigma}{2} \|\lambda^\star_\epsilon(\sigma; C)\|^2.$$

Asymptotic Optimality

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- Information gathered under the optimal sampling rule

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- Regularization penalty

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$$\lim_{\epsilon, \sigma \to 0} D^\star_\epsilon(\sigma; C) = D^\star(C).$$

- As $\epsilon, \sigma \to 0$, the modified hardness parameter converges to the optimal hardness $D^\star(C)$

Asymptotic Optimality

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$$\lim_{\epsilon, \sigma \to 0} D^\star_\epsilon(\sigma; C) = D^\star(C).$$

$$\frac{1}{D^\star(C)} \leq \liminf_{\delta \downarrow 0} \inf_{\pi \ \delta\text{-correct}} \frac{\mathbb{E}_C[\tau_\delta(\pi)]}{\ln(1/\delta)} \leq \limsup_{\delta \to 0} \frac{\mathbb{E}_C[\tau_\delta]}{\ln(1/\delta)} \leq \frac{1}{D^\star_\epsilon(\sigma; C)}.$$

Alternative Stopping Rule

$$Z(t) = \inf_{C' \in \textsc{Alt}(C_t)} \sum_{(i,j) \in I} N_{ij}(t) \, \mathrm{KL}((\widehat{v}_t)_{ij}, v'_{ij})$$

- No closed-form expression for $Z(t)$

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- No closed-form expression for $Z(t)$
- Workaround: "Factoring out" $\min_{(i',j') \in I} N_{i'j'}(t)$:

$$\widehat{Z}(t) = \left[\min_{(i',j') \in I} N_{i'j'}(t)\right]\left[\inf_{C' \in \textsc{Alt}(C_t)} \sum_{(i,j) \in I} \mathrm{KL}((\widehat{v}_t)_{ij}, v'_{ij})\right].$$

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- The associated (alternative) stopping rule is

$$\widehat{\tau}_\delta = \inf\{t \in \mathbb{N} : \widehat{Z}(t) > \beta(t, \delta)\}.$$

Theorem (Alternative Stopping Rule is δ-Correct)

Because $Z(t) \geq \widehat{Z}(t)$, the alternative stopping rule is still δ-correct.

Near-Optimality

Theorem (Near-Optimal Sample Complexity)

For any $\sigma > 0$, under D-tracking and the alternative stopping rule,

$$\limsup_{\delta \to 0} \frac{\mathbb{E}_C[\widehat{\tau}_\delta]}{\ln(1/\delta)} \leq \frac{1}{\tilde{D}^\star_\epsilon(\sigma; C)},$$

where

$$\tilde{D}^\star_\epsilon(\sigma; C) = \min_{(i,j)\in I} \big(\lambda^\star_\epsilon(\sigma; C)\big)_{ij} \inf_{C' \in \textsc{Alt}(C)} \sum_{(i,j)\in I} \mathrm{KL}(v_{ij}, v'_{ij}) - \frac{\sigma}{2} \|\lambda^\star_\epsilon(\sigma; C)\|^2.$$

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- Minimum sampling proportion

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- Overall KL separation from the alternatives

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- Small regularization penalty

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A^{3}CNP

Almost Asymptotically Optimal Active Clustering with Noisy Pairwise Observations

Suboptimality Gap

- We define the multiplicative suboptimality gap:

$$\mathrm{SG}_\epsilon(\sigma;C) \;=\; \frac{D^\star_\epsilon(\sigma;C)}{\widetilde{D}^\star_\epsilon(\sigma;C)} \;=\; \frac{1/\widetilde{D}^\star_\epsilon(\sigma;C)}{1/D^\star_\epsilon(\sigma;C)}$$

Suboptimality Gap

- We define the multiplicative suboptimality gap:

Upper bound under the alternative stopping rule

$$\mathrm{SG}_\epsilon(\sigma; C) = \frac{D^\star_\epsilon(\sigma; C)}{\widetilde{D}^\star_\epsilon(\sigma; C)} = \frac{\boxed{1/\widetilde{D}^\star_\epsilon(\sigma; C)}}{\boxed{1/D^\star_\epsilon(\sigma; C)}}$$

Upper bound from the GLR-based stopping rule

Suboptimality Gap

- We define the multiplicative suboptimality gap:

$$\mathrm{SG}_\epsilon(\sigma; C) = \frac{D^\star_\epsilon(\sigma; C)}{\widetilde{D}^\star_\epsilon(\sigma; C)} = \frac{1/\widetilde{D}^\star_\epsilon(\sigma; C)}{1/D^\star_\epsilon(\sigma; C)} \leq \frac{\overline{m}/\underline{m}}{\underbrace{1 - O_\epsilon(\sigma)}_{\approx 1}}$$

$$\overline{m} = \max_{(i,j)\in I} \left(\lambda^\star_\epsilon(\sigma; C)\right)_{ij}, \qquad \underline{m} = \min_{(i,j)\in I} \left(\lambda^\star_\epsilon(\sigma; C)\right)_{ij}.$$

Suboptimality Gap

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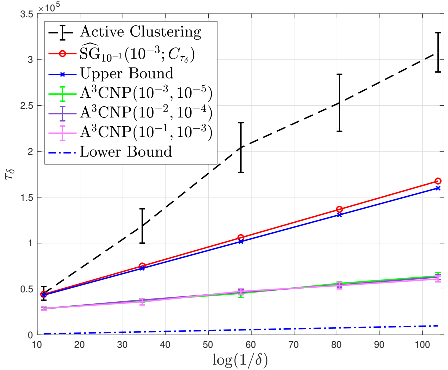
- We define the multiplicative suboptimality gap:

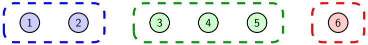
$$\mathrm{SG}_\epsilon(\sigma; C) = \frac{D^\star_\epsilon(\sigma; C)}{\widetilde{D}^\star_\epsilon(\sigma; C)} = \frac{1/\widetilde{D}^\star_\epsilon(\sigma; C)}{1/D^\star_\epsilon(\sigma; C)} \leq \frac{\overline{m}/\underline{m}}{\underbrace{1 - O_\epsilon(\sigma)}_{\approx 1}}$$

- In practice, we use the data-dependent proxy:

$$\widehat{\mathrm{SG}}_\epsilon(\sigma; C_t) = \frac{\max_{(i,j)\in I} \left(\lambda^\star_\epsilon(\sigma; C_t)\right)_{ij}}{\min_{(i,j)\in I} \left(\lambda^\star_\epsilon(\sigma; C_t)\right)_{ij}}.$$

Numerical Experiment





$$p = 0.6, \;\; q = 0.4$$

- $\mathrm{A}^3\mathrm{CNP}$ has slope close to the lower bound slope
- $\widehat{\mathrm{SG}}_\epsilon(\sigma; C_t)$ closely tracks the upper bound
- $\mathrm{A}^3\mathrm{CNP}$ significantly improves over the Active Clustering algorithm of Chen et al. (2023)[1] as $\delta \to 0$

[1] Y. Chen, R. K. Vinayak, and B. Hassibi, "Crowdsourced clustering via active querying: Practical algorithm with theoretical guarantees," *Proceedings of the AAAI Conference on Human Computation and Crowdsourcing*, vol. 11, no. 1, pp. 27–37, Nov. 2023. [Online]. Available: `https://ojs.aaai.org/index.php/HCOMP/article/view/27545`

Summary

Department of Artificial Intelligence

భారతీయ సాంకేతిక విజ్ఞాన సంస్థ హైదరాబాద్
भारतीय प्रौद्योगिकी संस्थान हैदराबाद
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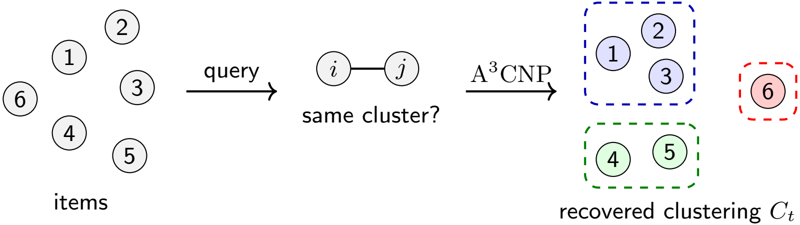
- We formulated noisy pairwise active clustering as a fixed-confidence bandit problem
- We derived an instance-dependent lower bound on the expected sample complexity
- We proposed $\mathrm{A}^3\mathrm{CNP}$ (D-tracking + efficient stopping rule), and proved it is near-optimal



Full paper available at arXiv:2602.05690